

Title: Modos Paper Monitor

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Markdown Content: Modos Tech

E-Paper

FPGA Boards

A fast, low-latency, open-hardware e-paper monitor and dev kit

Video 9



\$197,588 raised

of \$110,000 goal

179% Funded! Order Below

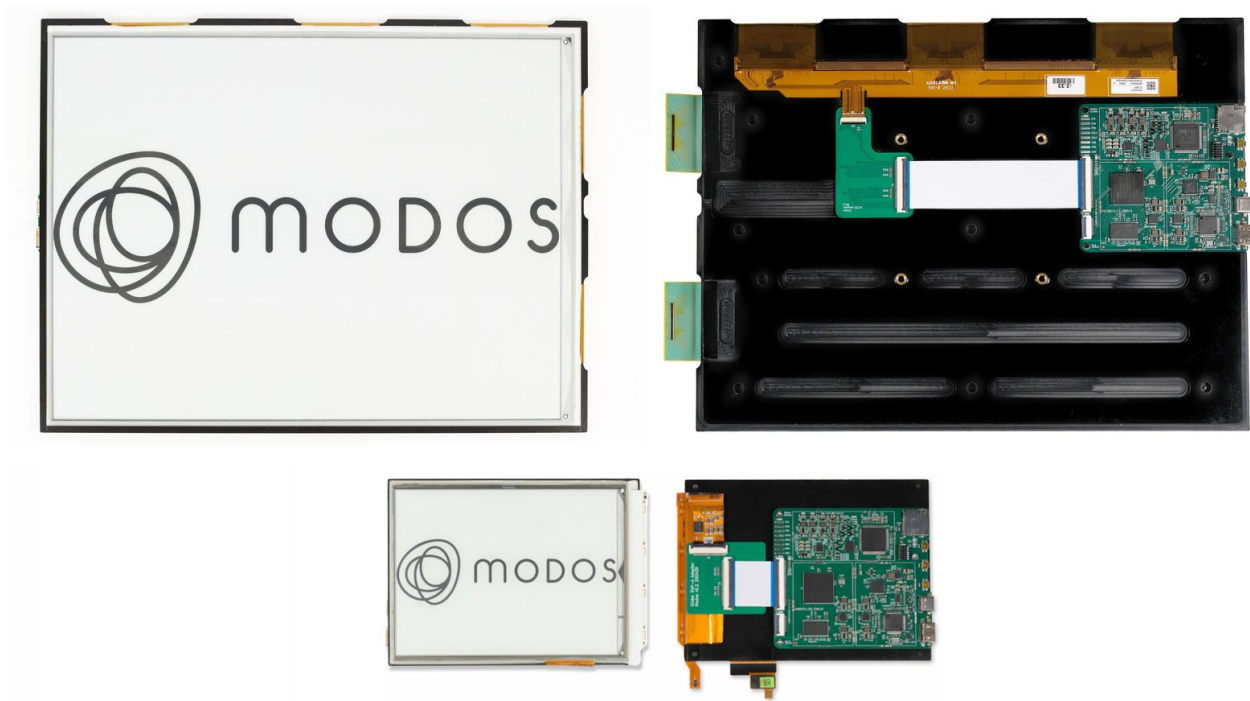
In stock. Order now, ships within three business days.

\$249 - \$699 View Purchasing Options

Recent Updates

- Apr 01, 2026 Progress on Shipping, Cases, and Firmware
- Feb 04, 2026 First Shipment Starting Soon
- Dec 08, 2025 Highlights, Certifications, and Production Progress
- Sep 17, 2025 A Beginning, Not an End

Modos Paper Dev Kit includes nearly everything you need to create an open-hardware e-paper monitor with a fast 75 Hz refresh rate, low latency, various screen-update configurations, multiple image modes, and flexible dithering options. It can be connected through HDMI or USB, and it works on Linux, macOS, and Windows. We are offering 6-inch and 13-inch monochrome kits, both powered by the same display controller, which can also be used with other panels.



At the Cutting Edge of E-Paper

Twenty years ago, the release of the first e-paper devices marked a significant milestone in display technology, giving rise to e-readers and digital signage. Electrophoretic displays stand out for their unique qualities: they are easy on the eyes, mimic the aesthetics of paper, conserve power when idle, and offer high contrast with wide viewing angles even in direct sunlight.

In recent years, improvements in e-paper technology have expanded its applications beyond traditional e-readers. Despite these advancements, however, key challenges still hold back widespread adoption of e-paper devices:

- **Proprietary Hardware and Software:** Most e-paper devices operate on closed, proprietary systems.
- **Lack of Standards:** There is a shortage of best practices and guidelines for designing user interfaces optimized for e-paper.
- **High Costs:** The expense of e-paper panels limits experimentation and broader adoption.

We built the Modos Developer Kit to allow engineers, product designers, programmers, and enthusiasts to re-purpose e-paper screens in creative ways. We're also building a community that's committed to establishing the standards, best practices, and guidelines needed to realize the full potential of e-paper technology. Join us on Discord and by following us on Mastodon, Bluesky, Matrix, and Twitter.

High Refresh Rate

Video 10

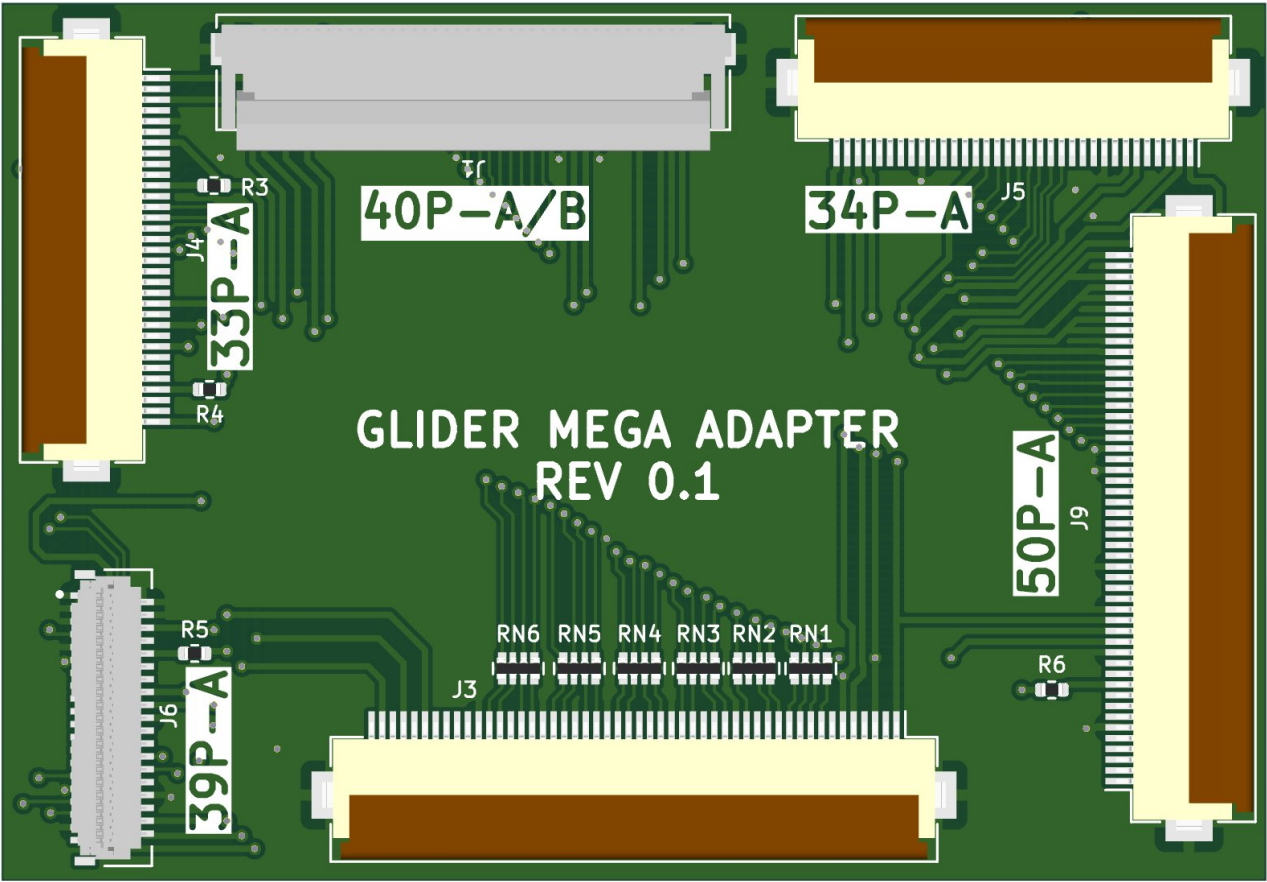
Our Developer Kit leverages Caster, an open-source FPGA-based electrophoretic display controller engineered for low-latency performance and capable of driving e-paper screens at up to 60 Hz. By dividing the screen into multiple update regions, the Caster processes and displays new images or text instantly, without waiting for previous updates to complete. Each pixel is managed independently, and early cancellation techniques ensure that pixels update as soon as new data is available. The result is smooth, responsive performance with high frame rates and superior contrast, ideal for dynamic content and fluid interactions.

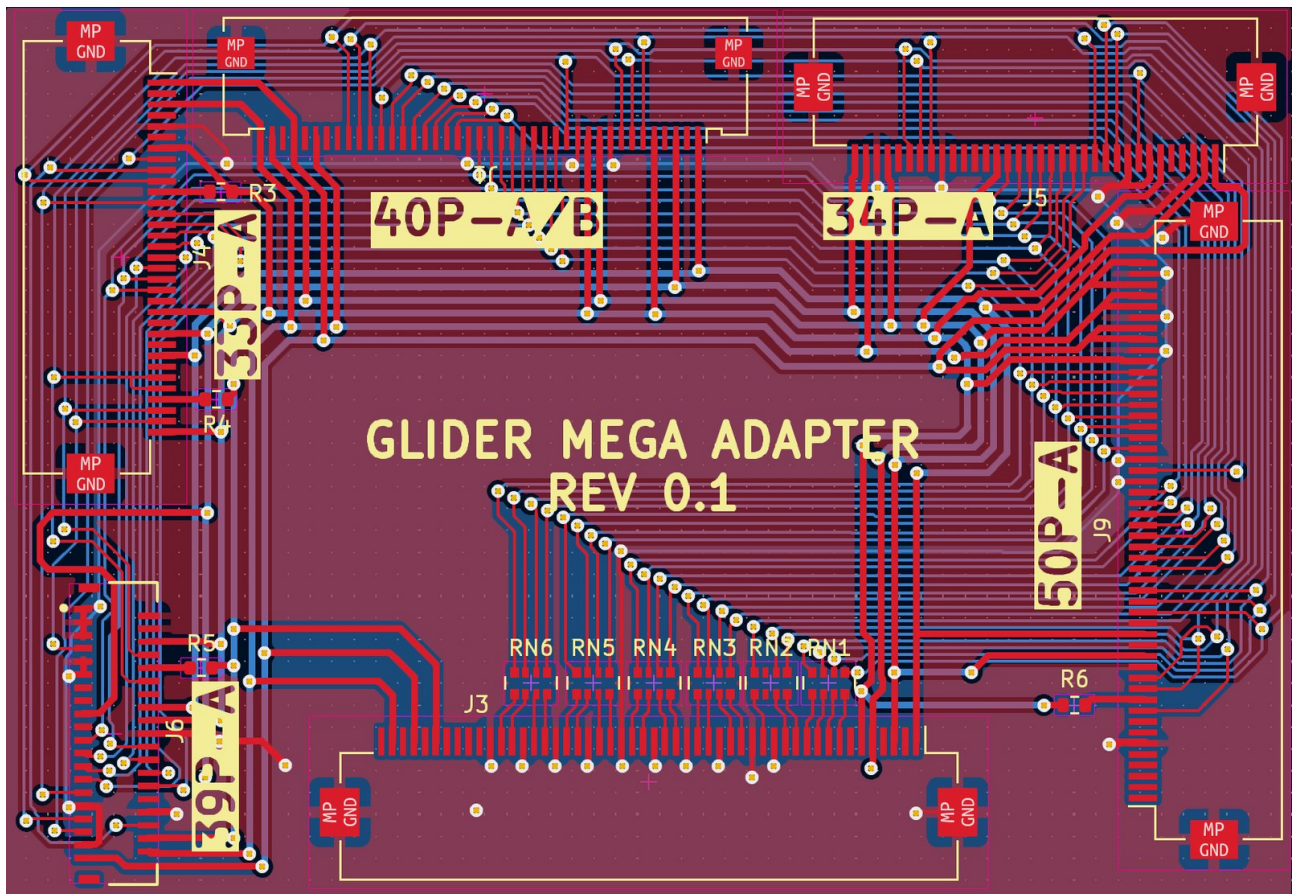
Compatible With a Variety of Screens



Got a spare e-paper screen gathering dust or looking to re-purpose a display for your project? Our Developer Kit makes it simple. Compatible with a wide range of screens—from 6-inch up to 13.3-inch, in both monochrome and color—the kit lets you easily connect and reuse unused displays. With our monitor enclosure design files, you can build a custom monitor housing that perfectly fits your project's needs.

Explore our [GitHub repository](#) for a complete list of compatible screens.





Every kit includes the **Glider Mega Adapter**, giving you out-of-the-box support for a wide range of e-paper panels. With one adapter set, you can **connect displays from 4.3” up to 13.3”**. The Glider Mega Adapter includes support for the following (which is not a comprehensive list, as some panels are untested but expected to work):

- **39-pin 0.3mm pitch:** Certain 4.3”, 5”, and 13.3” screens (ED043WC1, ED043WC3, ED050SU3, ES133UTx, ED133UTx, EC133UJ1)
- **33-pin 0.5mm pitch:** Most 9.7” screens (ED097OCx, ED097ODx, ED097TCx)
- **0.5mm pitch, 7.8”–10.3” screens:** (ED078KCx, ED078KHx, EC078KHx, ED080TC1, EC080SC2, ED100UC1, ES103TCx, ED103TCx)
- **34-pin 0.5mm pitch:** Most 6.0” screens (ED060SCF, ED060SCP, ED060XC3/C5/C8/C9/CG/D4/D6/G1/H2, ED060KC1/C4/D1/G1, etc.)
- **50-pin 0.5mm pitch:** Most 11.3” screens (ED113TCx)

With these adapters, anyone interested in working with e-paper has much more flexibility to experiment by reusing e-paper panels from existing devices as well as trying out new ones.

User-Defined Modes & API

Video 11

While commercial e-paper products rely on preset driving strategies (text, graphics, video modes) to cope with the technology’s limitations, our Developer Kit goes further. It offers full customization so you can tailor the display experience to your specific needs.

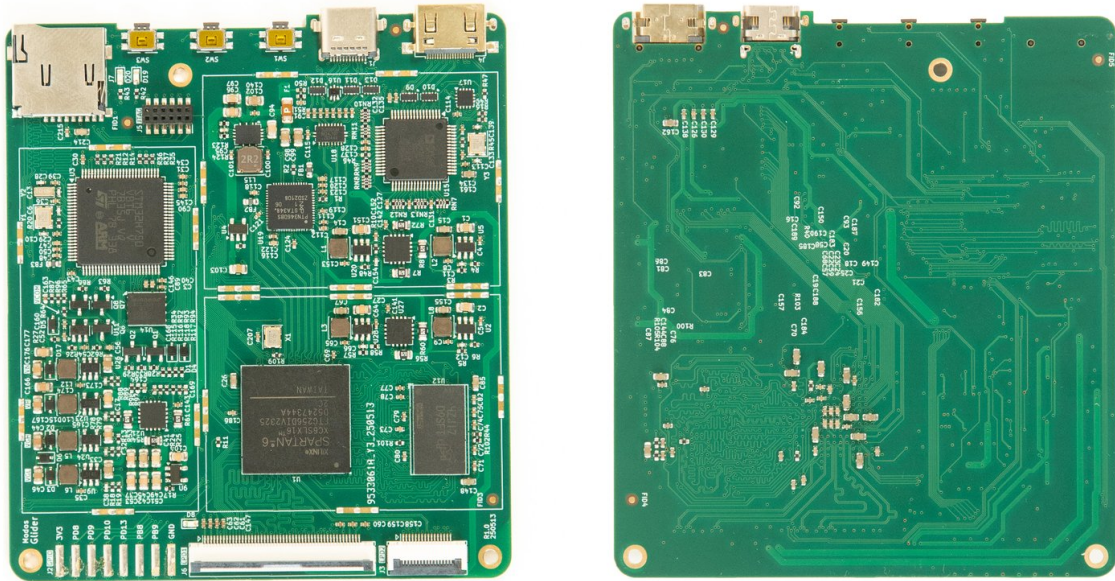
For example, the Hybrid Greyscale Mode allows the Caster to switch between a fast binary mode for quick updates and a slower, detailed greyscale mode for refined rendering. This minimizes latency and visual flashing by updating in binary first, then refining in greyscale as the image stabilizes.

With our C API, you gain direct control over screen and mode selection, letting you optimize the display for a wide range of applications. Our broader vision includes developing native e-paper applications and a protocol for seamless mode transitions based on content type.

Features & Specifications

Driver Board Hardware Reference Design

- Xilinx Spartan-6 LX16 FPGA running Caster gateway
- DDR3-800 framebuffer memory
- Video input supporting:
 - USB Type-C DisplayPort Alt-Mode with on-board PTN3460 decoder
 - DVI (via microHDMI connector) video input with on-board ADV7611 decoder
- E-paper power supply with up to 1 A peak current on the ± 15 V rail to support large panel sizes.
- VCOM kick-back voltage measurement support
- On-board STM32H750 microcontroller for USB communication, firmware upgrades, and standalone applications. Processing rate up to 133 MP/s when error-diffusion dithering enabled and 200 MP/s when disabled



Caster FPGA Gateway Reference Design

- Supports electrophoretic display panels with parallel interfaces (E Ink, OED, and DES)
- Compatible with both monochrome and color-filter-array screens
- Extremely low processing delay (< 20 μ s)
- Supports binary, 4-level grayscale, and 16-level grayscale output modes
- Latency-optimized binary and 4-level grayscale driving modes
- Hybrid automatic binary and 16-level grayscale driving mode
- Host software runtime controllable regional updates and mode switching
- Hardware bayer dithering, blue-noise dithering, and error-diffusion dithering with no additional latency

Comparisons

E-Paper Monitors

	Modos 13" Paper Dev Kit	DASUNG 13.3" Paperlike HD-FT	Boox Mira 13.3" E-Ink Monitor	Waveshare EINK DISP 133
Display	Eink Carta 1000 (ED133UT3)	Eink Carta 1000 (ES133TT3)	Eink Carta 1200 (ES133TT5)	Eink Carta 1000 (ED133UT2)
Screen Size	13.3"	13.3"	13.3"	13.3"

	Modos 13" Paper Dev Kit	DASUNG 13.3" Paperlike HD-FT	Boox Mira 13.3" E-Ink Monitor	Waveshare EINK DISP 133
Screen Resolution	1600 x 1200	2200 x 1650	2200 x 1650	1600 x 1200
Maximum Frame rate	75Hz	40Hz	~20Hz	~15Hz
Interfaces	HDMI & USB Type-C	HDMI	HDMI & USB Type-C	HDMI
Touchscreen	No	Yes	No	No
Frontlight	No	Yes	Yes	No
Open Hardware	Yes	No	No	No
Programmable	Yes	No	No	No
Modes				
API for Display Controller	Yes	No	No	No
Price	\$599	\$798	\$800	\$679

E-Paper Dev Kits

	Modos 6" Paper Dev Kit	Waveshare e-paper HAT	Inkplate 6 MOTION	EPDiY
Controller	FPGA-based	None	ESP32-based	ESP32-based
Display Technology	E Ink	E Ink	E Ink	E Ink
Screen Size	6.0"	6.0"	6.0"	9.7"
Programming Language	Rust, Python, C	Python, C, C++	MicroPython, C, C++	MicroPython, C, C++
Support				
Screen Resolution	1448 x 1072	1448 x 1072	1024 × 758	1200 x 825
Maximum Frame rate	75 Hz	< 2 Hz	11 Hz	Unknown
Interfaces	HDMI, USB Type-C	SPI, GPIO	USB Type-C, GPIO	GPIO
Touchscreen	No	No	No	N/A
Frontlight	No	No	Yes	N/A
Open Hardware	Yes	No	Yes	Yes
Programmable	Yes	Yes	Yes	Yes
Modes				
API for Display Controller	Yes	No	No	No
Works With Other Panels	Yes	No	No	Yes
Power Consumption	High	Low	Low	Low
Price	\$199	\$113	\$195	N/A

Support & Documentation

The Modos Developer Kit is open hardware built using open-source software, with all design files and source code available on GitHub. Our documentation offers extensive information on electrophoretic displays, drawing from both public sources and our original research. This serves as a comprehensive guide for anyone looking to dive deeper into e-paper technology.

For additional support or to join the discussion, please visit Discord and follow us on Mastodon, Bluesky, Matrix, and Twitter.

Thank You to the NLnet Foundation and the NGI Zero Entrust Fund

Our team would like to acknowledge partial funding for development of Caster from the NGI0 Entrust Fund, a fund established by NLnet with financial support from the European Commission's Next Generation Internet program.



In the Press







Produced by Modos Tech in Boston, MA.

Sold and shipped by Crowd Supply.

6" Modos Paper Dev Kit

Pre-assembled development kit for the 6-inch version of our low-latency, 1448 x 1072 e-paper display. Includes a screen, a mainboard, a ribbon cable, and a Glider Mega Adapter to support e-paper panels of various sizes.

\$249 \$8 US Shipping / \$18 Worldwide

In stock

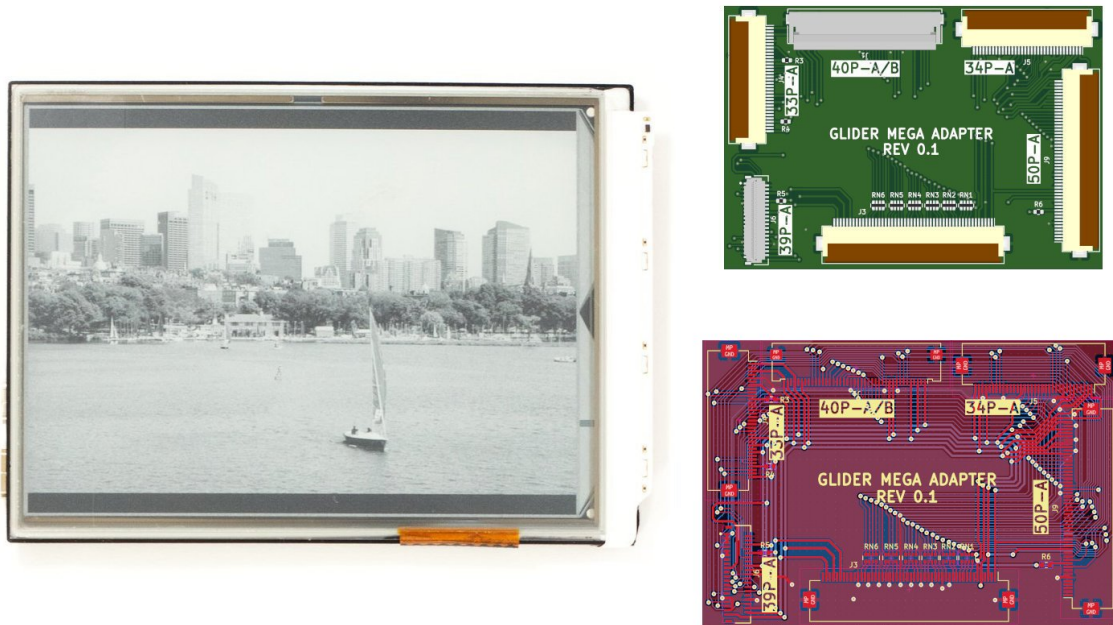


Figure 1: Image 25: 6" Modos Paper Dev Kit

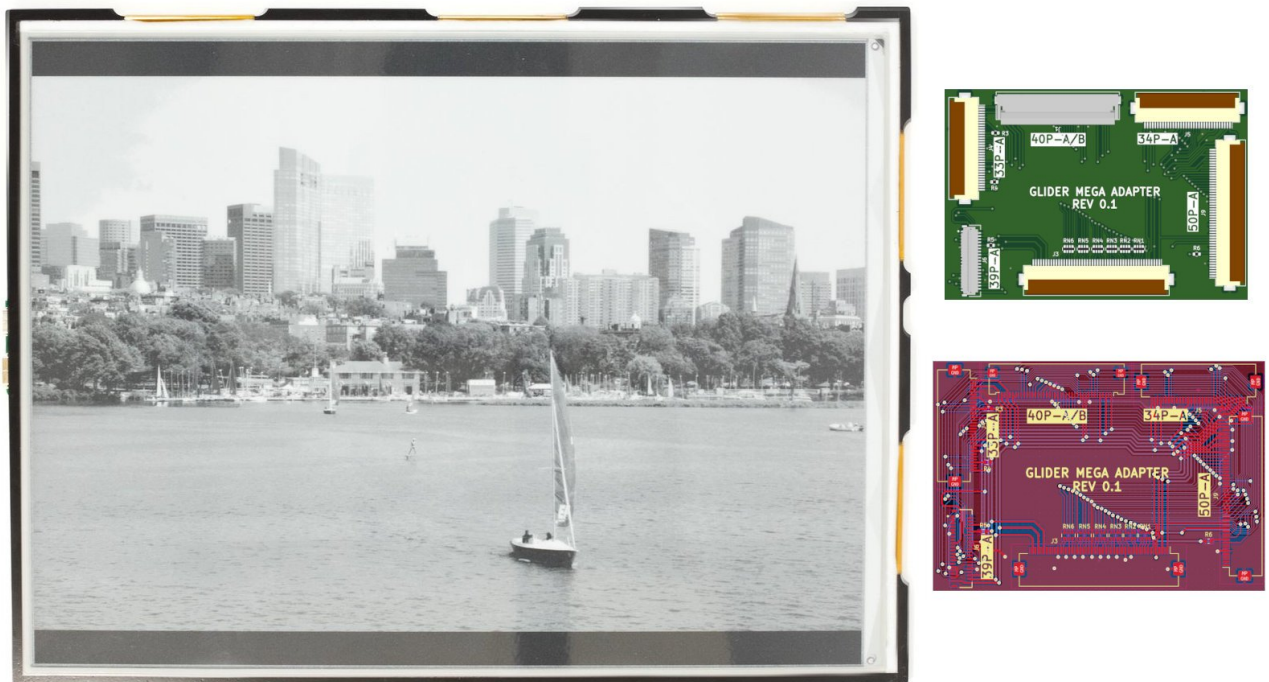


Figure 2: Image 26: 13" Modos Paper Dev Kit

13" Modos Paper Dev Kit

Pre-assembled development kit for the 13-inch version of our low-latency, 1600 x 1200 e-paper display. Includes a screen, a mainboard, a ribbon cable, and a Glider Mega Adapter to support e-paper panels of various sizes.

\$699 Free US Shipping / \$18 Worldwide

In stock

About the Team



Figure 3: Image 27: Modos Tech



Figure 4: Image 28

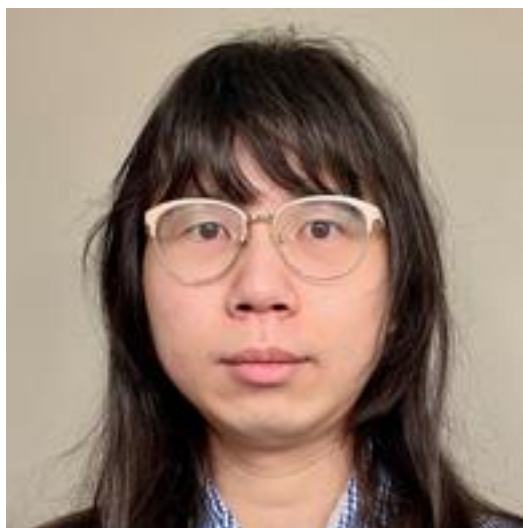


Figure 5: Image 29